Mechanical Work and Power

Notes: concepts, equations and worked interpretation

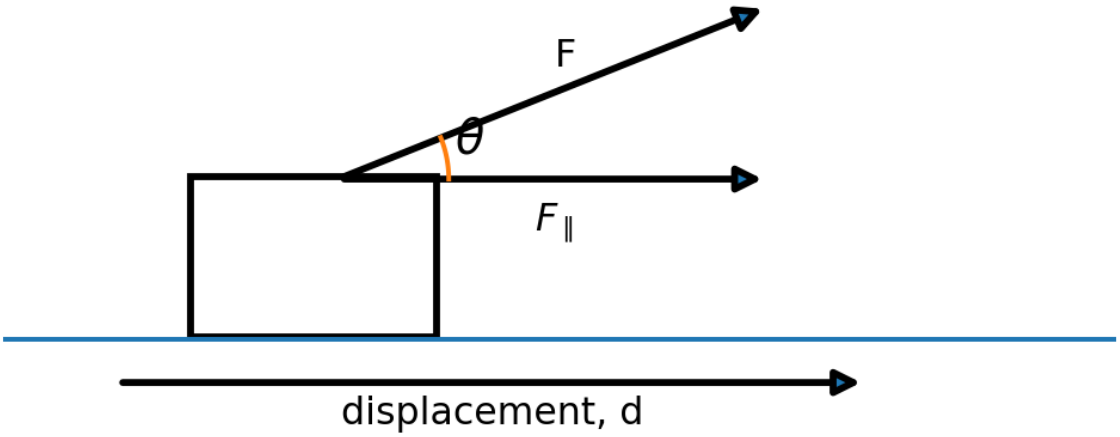
These notes use the current AP Physics 1 conventions and equation-sheet notation. Numerical examples use the sequence: equation → rearrange → substitute SI values → answer with units.

Lesson goals. Describe work done by individual forces and by the net force; distinguish positive, negative and zero work; use force-displacement graphs; apply the work-energy theorem; distinguish conservative and nonconservative forces; and describe energy transfer using average and instantaneous power.

1. What does work mean in AP Physics?

Mechanical work is a way energy is transferred into or out of a selected system by a force acting through a displacement. Work is a scalar and is measured in joules (J). The key question is not simply “is there a force?” but “does the point of application of that force move, and what component of the force is parallel to that displacement?”

$$W = F_{\parallel} d = Fd\cos\theta$$



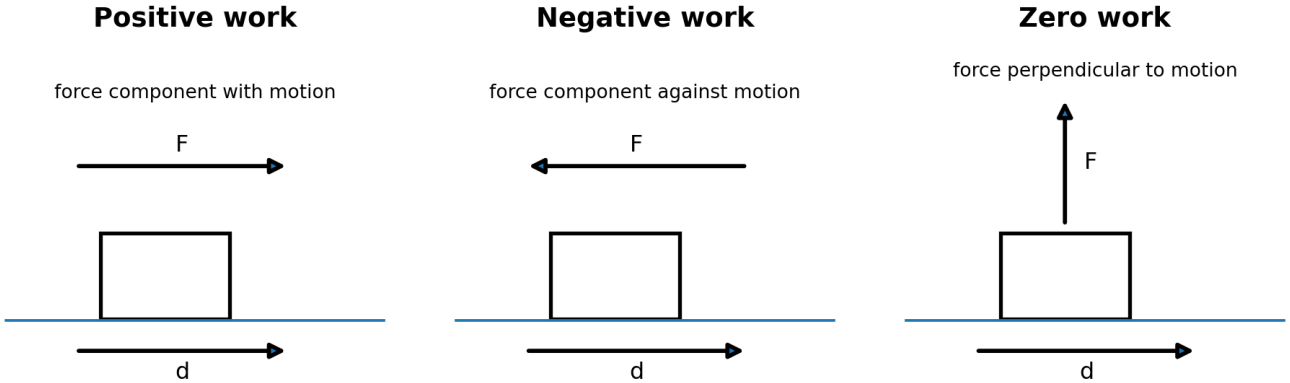
For a constant force:

$$\mathbf{W = F_{\parallel} d = Fd\cos\theta}$$

Here θ is the angle between the force and the displacement. $F_{\parallel}$ is the component of the force parallel to the displacement. If there is no displacement, the work done by that force is zero.

Important: work is done by a particular force on an object or system. Different forces can do positive, negative or zero work at the same time.

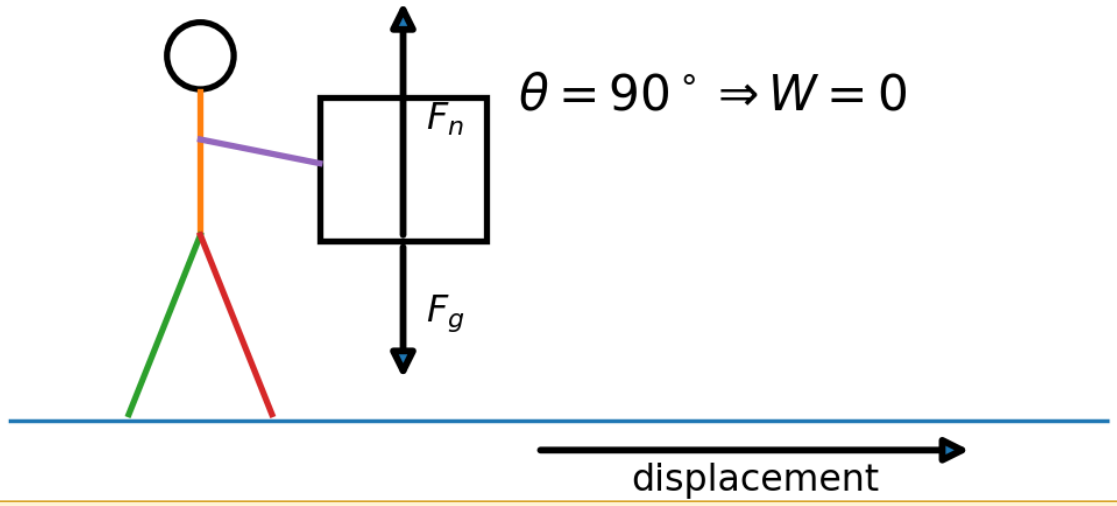
2. Positive, negative and zero work



Positive work: the force has a component in the direction of displacement ($\cos\theta > 0$). Positive work by one force transfers energy to the selected system.

Negative work: the force has a component opposite the displacement ($\cos\theta < 0$). Kinetic friction commonly does negative work on a moving object.

Zero work: the force is perpendicular to the displacement ($\theta = 90°$), or the point of application does not move. A person carrying a box horizontally at constant height is a standard example: the upward support force on the box is perpendicular to its horizontal displacement, so that force does zero mechanical work on the box.



Do not overinterpret one force. Positive work by one force does not guarantee that an object speeds up. The object speeds up only if the net work is positive.

3. Net work and the work-energy theorem

The work-energy theorem relates the sum of the work done by all forces to the change in an object's translational kinetic energy:

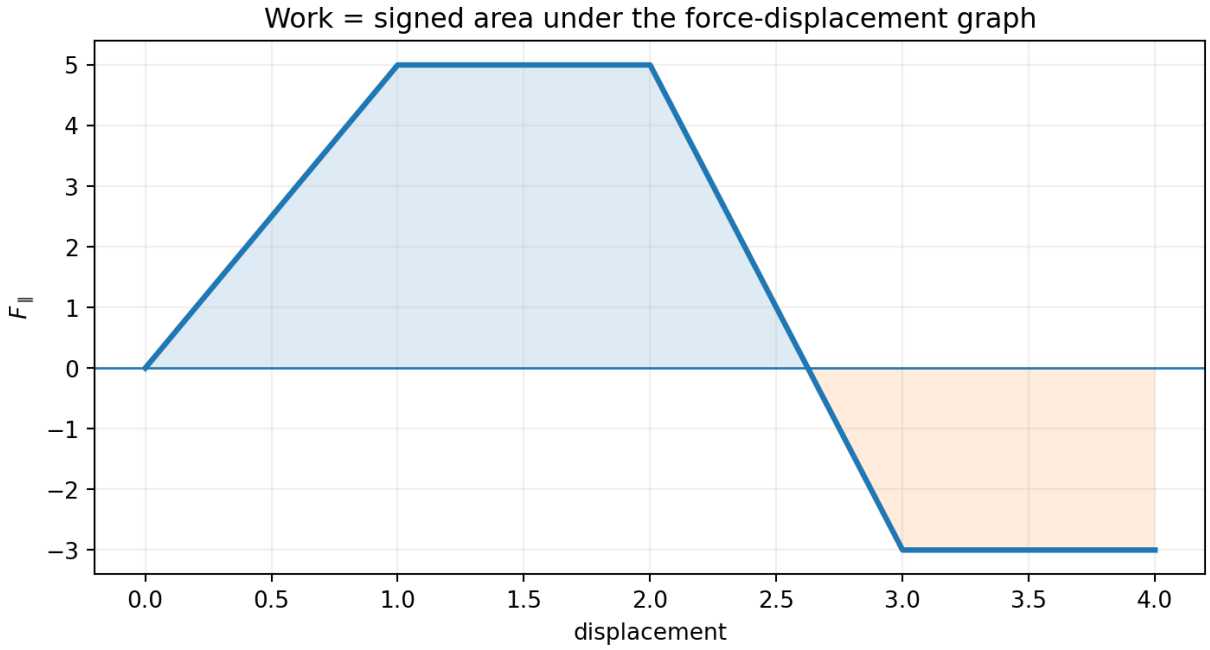
$$\Delta K = \Sigma W_i = \Sigma F_{\parallel,i} d_i$$

Therefore: net work > 0 → K increases; net work < 0 → K decreases; net work = 0 → K is unchanged. This is about kinetic energy, not necessarily speed direction.

The theorem is especially useful when several forces act and you can determine the work done by each force, or when the net force is given as a function of position.

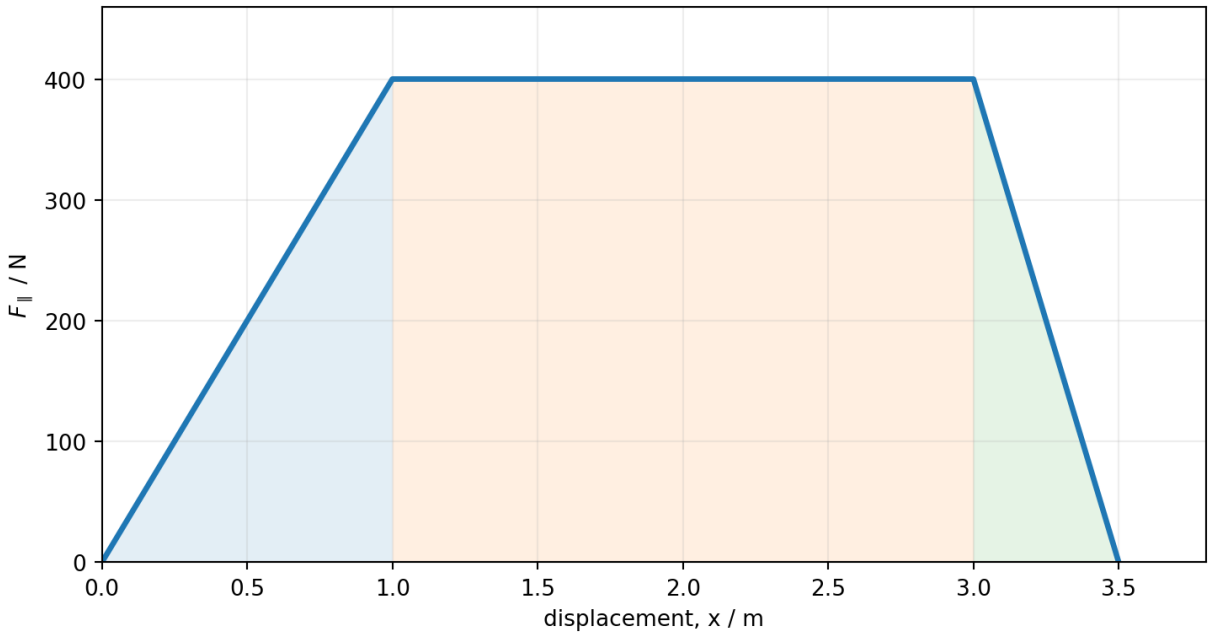
4. Variable forces and force-displacement graphs

If $F_{\parallel}$ changes with position, do not use one constant value in $W = F_{\parallel} d$. The work done equals the signed area between the $F_{\parallel}$-versus-displacement graph and the displacement axis.



Area above the axis contributes positive work; area below the axis contributes negative work. For piecewise-linear graphs, split the area into rectangles, triangles and trapezoids.

Your worksheet graph. The force rises from 0 to 400 N over the first 1.0 m, remains at 400 N from 1.0 m to 3.0 m, and then falls to zero over the final 0.50 m. The total area is 1100 J.



5. Conservative and nonconservative forces

Conservative force: the work depends only on the initial and final configuration, not on the path. Potential energy is associated with conservative interactions. Over a closed path, the net work done by a conservative force is zero.

Nonconservative force: the work can depend on the path. Friction and air resistance are standard examples. For kinetic friction acting opposite the displacement, the work on the object is $W_f = -F_f d$. The magnitude $F_f d$ is often used to describe mechanical energy dissipated to internal/thermal energy.

Mechanical energy is the sum of a system's kinetic and potential energies. Whether mechanical energy is constant depends on the system chosen and on transfers across its boundary.

6. Power: how fast energy is transferred

Power describes the rate of energy transfer or conversion. The SI unit is the watt: $1\ \text{W} = 1\ \text{J s}^{-1}$.

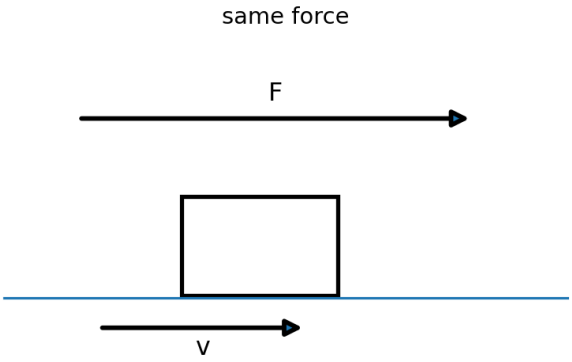
$$\mathbf{P_{avg} = W/\Delta t = \Delta E/\Delta t}$$

Average power describes the total work or energy change over a finite time interval.

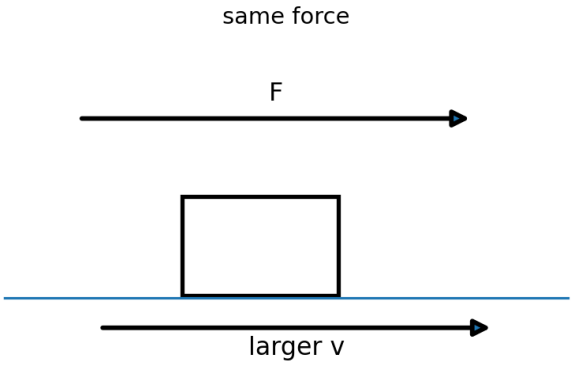
$$\mathbf{P_{inst} = F_{\parallel} v = Fv\cos\theta}$$

Instantaneous power depends on the component of the force parallel to the velocity and on the speed at that instant. A constant force does not imply constant power if the speed is changing.

Lower instantaneous power



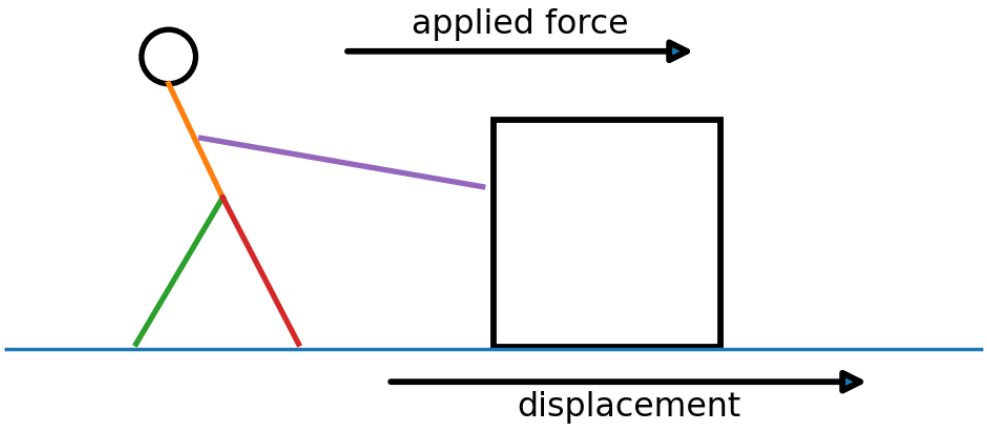
Higher instantaneous power



Power can also be read from the slope of an energy-versus-time or work-versus-time graph. A steeper slope means a larger magnitude of instantaneous power.

7. Worked interpretation: the 480 N box

A person pushes a 360 kg box horizontally with a constant 480 N force through 3.5 m. With no friction, the applied force is the net horizontal force.



Work by the person: $W = Fd = (480)(3.5) = 1680$ J. The net work is also 1680 J, so $\Delta K = +1680$ J.

If the box speeds up, $P_{inst} = Fv$ increases even though F is constant. Therefore the person's instantaneous mechanical power is not constant.

8. Optional extension from the original worksheet: efficiency

Efficiency is useful in many energy problems, but it is not listed as required Unit 3 content or an AP Physics 1 equation-sheet relationship in the current course description. It is therefore best treated here as an extension:

efficiency = useful energy output / total energy input = useful power output / total power input

Multiply by 100 to express the efficiency as a percentage.

9. Equation checklist

Situation	Equation / relationship	Meaning
Constant force	$W = F_{\parallel} d = Fd\cos\theta$	Work done by one force
All forces	$\Delta K = \Sigma W_i$	Net work changes kinetic energy
Variable force	W = signed area under $F_{\parallel}$ vs displacement	Graphical work
Average power	$P_{avg} = W/\Delta t = \Delta E/\Delta t$	Energy transfer per time interval
Instantaneous power	$P_{inst} = F_{\parallel} v = Fv\cos\theta$	Power at an instant